RANSOMWARE

ABDURRAHMAN QURESHI

242466



What is Ransomware?

Malware that encrypts files and demands payment for release

Comes from words "ransom" and "software" combined

It holds your data hostage, not just steals it

Payment demanded in cryptocurrency, which is hard to trace

Targets individuals, hospitals, banks, schools, and governments

Like someone changing all your locks and selling keys



How It WORKS

Delivery : phishing email, malicious link, fake download, USB drive

Infection : malicious code runs, may stay silent for days

Encryption : files locked using AES and RSA encryption

Ransom Note : payment instructions, deadline, and amount displayed

Extortion : data often stolen first, then payment demanded



Types of RANSOMWARE

Crypto Ransomware : encrypts files, most common type today

Locker Ransomware : locks you out of entire device

Scareware : fake warnings, usually no real encryption

Leakware : steals data, encrypts it, threatens to publish



Motives to ATTACK

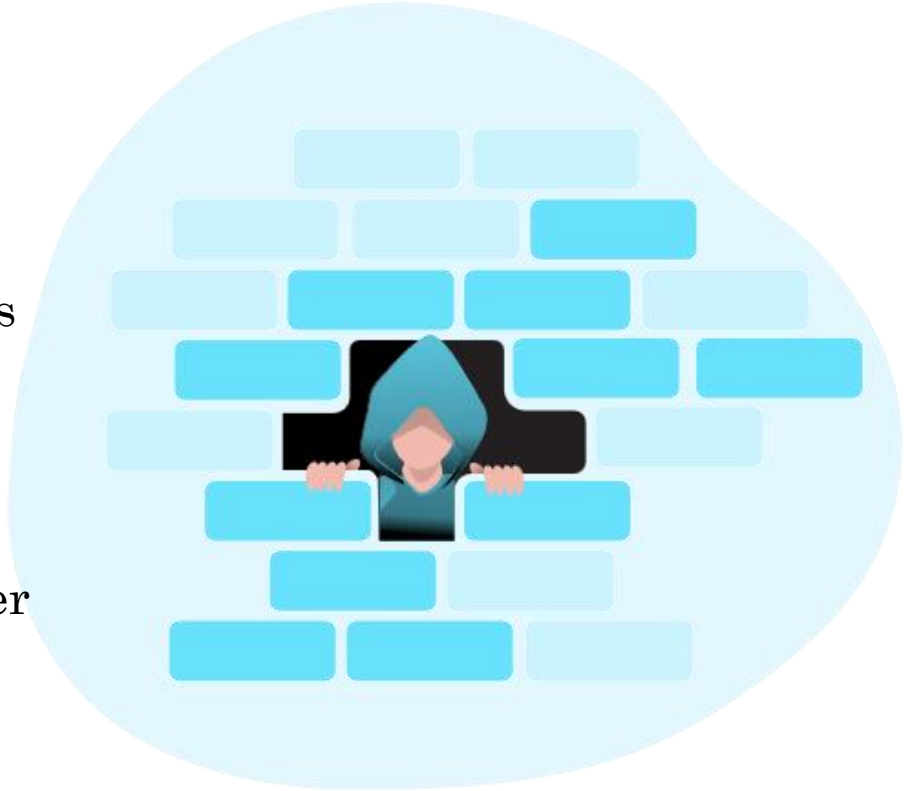
Money : primary driver, millions paid in cryptocurrency

RaaS : kits sold to criminals, profits shared between them

High-value targets : hospitals, banks, schools, governments targeted

Geopolitics : some groups protected or tolerated by states

Low risk, high reward : anonymous payments, cross-border attacks



FAMOUS Attack INSTANCES

WannaCry (2017)

- Hit 200,000+ computers across 150 countries worldwide
- Crippled UK's NHS hospitals, surgeries and appointments cancelled

NotPetya (2017)

- Disguised as ransomware and caused approximately $10 billion in global damage

Colonial Pipeline (2021)

- US fuel pipeline shut down for several days
- ~$4.4M ransom paid, gas shortages across East Coast

AIIMS Delhi (2022)

- Hospital services disrupted for days across India



IMPACT

Data loss : files permanently gone without proper backups

Downtime : business operations completely halted for days

Financial loss : ransom plus recovery and legal costs

Reputation damage : lost customer trust and brand value

Real-world harm : delayed surgeries, fuel shortages, closed businesses



PREVENTION & PROTECTION

Backups : offline, offsite, tested regularly, best defense

Never open **unknown** links or email attachments

Keep systems **updated** and **patched** against vulnerabilities

Use antivirus, tools, and firewalls

Use strong password

Never pay ransom : no guarantee, funds more crime

